

Hybrid Quant Engine

External API Budget Estimate & Requirements

Field	Value
Document Version	1.0
Date	13 July 2026
Prepared For	Client Approval — API & Data Vendor Budget
Project	Hybrid Quant Engine (Analysis, Backtest & Signals)
Markets	India (NSE/BSE) + US (NYSE/NASDAQ)
Scope	Research & signals only — no buy/sell or broker integration
Currency	INR (₹); USD converted at ₹84 / \$1 where applicable
GST	18% applicable on Indian vendor invoices (shown separately)

1. Executive Summary

This document lists all external APIs required to build the Hybrid Quant Engine, including each vendor name, purpose, technical requirements, and estimated cost. Costs are presented in two scenarios: MVP (development / pilot) and Production (client-facing platform). Alert/notification APIs and optional infrastructure APIs are excluded from this budget.

Scenario	Monthly (₹)	One-Time Year 1 (₹)	Year 1 Total incl. GST (₹)
MVP — Low estimate	₹45,872	₹50,000	₹7.1 Lakh
MVP — High estimate	₹83,836	₹1,50,000	₹13.6 Lakh
Production — Low estimate	₹95,916	₹1,50,000	₹15.3 Lakh
Production — High estimate	₹2,25,000	₹4,00,000	₹36.6 Lakh

Recommendation for client approval: Begin with MVP Low (~₹7.1 Lakh Year 1 incl. GST) using a 50-symbol pilot universe. Scale to Production after backtest validation.

2. Project Assumptions

- 10-Point Confluence Scoring Matrix (fundamental + technical + order-flow).
- 20-year historical backtest window (2006–2026) with 0.25% simulated friction.
- Live market data during India shift (IST 09:15–15:30) and US shift (IST 19:00–02:30).
- Portal publishes signals/alerts only — no order execution.
- Backend: Laravel; AI: Gemini + RAG; DB: MySQL + MongoDB.
- TrueData API pricing is estimated (custom quote required from vendor).
- Polygon.io and FMP prices per published plans (July 2026).

3. Detailed API Breakdown by Category

Category A — India Market Data

Purpose: Live streaming, EOD history, delivery %, corporate-action-adjusted OHLCV for NSE/BSE.

#	API / Vendor	Purpose	Requirements	MVP Monthly (₹)	One-Time (₹)
A1	TrueData Market Data API (WebSocket + REST)	Live India shift streaming; real-time 5min bars; REST historical EOD; DelVolume; bhavcopy	NSE EQ + BSE EQ; ~50 (MVP) or ~200 (Prod) symbols; WebSocket port 8082; 5min bar stream enabled; REST auth via auth.truedata.in / history.truedata.in; corporate-action-adjusted EOD; delivery data for P9	8,000 – 20,000	—
A2	TrueData getBhavCopy (EQ / BSEEQ)	Bulk daily EOD + delivery post-market ingest	Segment EQ + BSEEQ; daily automated pull after 18:45 IST; store in MongoDB	Included in A1	—
A3	TrueData	20-year	Interval EOD	Included in	50,000 –

	getBarData (EOD / DelVolume)	backfill and ongoing daily bars	+ DelVolume; date range 2006-2026; split/bonus adjusted series confirmed with vendor	A1	1,50,000 (MVP backfill) 1,50,000 – 4,00,000 (Prod backfill)
A4	NSE / BSE Exchange Data Fees	Mandatory exchange licensing on top of TrueData	Non-professional or commercial license per use case; separate from TrueData subscription	2,000 – 5,000	—
A5	Exchange Redistribution License (Production only)	Display live/delayed prices on client-facing dashboard	Written NSE/BSE approval + TrueData redistribution agreement; required if end-users see live data	0 (MVP internal) 15,000 – 50,000 (Production)	—

Category A MVP subtotal: ₹10,000 – ₹25,000/month + ₹50,000 – ₹1,50,000 one-time backfill.

Vendor: <https://www.truedata.in/market-data-apis> | Quote: sales@truedata.in

Category B — US Market Data

Purpose: Live US shift streaming, 20Y EOD history, block/large trade data for order-flow scoring (P9).

#	API / Vendor	Purpose	Requirements	MVP Monthly (₹)	One-Time (₹)
B1	Polygon.io Stocks Advanced Plan	Real-time US WebSocket (Shift 2); 20+ years daily OHLCV; trades for block-trade analysis	NYSE + NASDAQ equities; WebSocket wss://socket.polygon.io; REST api.polygon.i	\$199 / ₹16,716	—

			o; unlimited API calls; real-time (non-professional tier); ~50-200 symbol universe		
B2	Polygon REST Aggregates / Trades	Historical and live bar construction; volume breakout (P9)	Daily + minute aggregates; trades endpoint for large/block trade ratio; JSON format	Included in B1	—
B3	Polygon Corporate Actions	Split/dividend adjustment for US backtest	Historical corp actions per symbol; point-in-time correctness for 20Y walk-forward	Included in B1	—
B4	US Exchange Pro Fees (if applicable)	Required if user classified as professional by exchange	Polygon Non-Pro vs Pro classification; additional fees may apply for commercial redistribution	0 – 8,400 (Production)	—

Category B subtotal: ₹16,716 – ₹25,116/month.

Vendor: <https://polygon.io/pricing>

Category C — Index & Benchmark Data

Purpose: Mansfield Relative Strength (P7) vs Nifty 50 and S&P 500.

#	API / Vendor	Purpose	Requirements	Monthly (₹)	One-Time (₹)
C1	TrueData — NIFTY 50	India benchmark index history + live	Symbol NIFTY 50; 20Y EOD + live stream during India	Included in A1	—

			shift; same adjustment standard as equities		
C2	Polygon.io — S&P 500 index	US benchmark index history + live	Index ticker via Polygon; 20Y daily; live during US shift	Included in B1	—

Category C subtotal: ₹0 (bundled in Categories A and B).

Category D — Fundamental & Financial Statement Data

Purpose: Scoring points P1–P5 (ROE, ROIC, Altman Z, Beneish M, Sloan, DSO, CAGR, PEG, FCFF/FCFE).

#	API / Vendor	Purpose	Requirements	MVP Monthly (₹)	One-Time (₹)
D1	TrueData Corporate REST API	India fundamentals, financial statements, ratios	corporate.getPnLById, getBalanceSheetById, getCashFlowDetailById; getAllResultsByCompany (quarterly/yearly); standalone + consolidated; quarterly refresh post-results	8,000 – 15,000	—
D2	TrueData Corporate WebSocket (optional)	Live corporate announcements during market hours	rtCorpConnect port 9092; announcement feed for RAG + alert context	2,000 – 5,000 (optional)	—
D3	Financial Modeling Prep (FMP) Premium	US fundamentals — income statement, balance sheet, cash flow, ratios	Premium plan: 750 calls/min; 30 years history; full fundamentals; US	\$59 / ₹4,956	—

			coverage; commercial use terms reviewed		
D4	FMP Ultimate (Production upgrade)	US transcripts, 13F, bulk delivery, higher rate limits	3,000 calls/min; bulk endpoints; needed at 500+ symbol scale	\$149 / ₹12,516 (Production)	—
D5	SimFin Pro API	US point-in- time fundamental s for honest 20Y backtest	20+ years history; 30,000 credits/mont h; avoids lookahead bias in P1–P5 backtest; annual billing \$71/mo equivalent	\$71 / ₹5,964 (recommend ed)	—
D6	TrueData getCorpActio n	India corporate actions for price adjustment	historical.get CorpAction per symbol; splits, bonuses, dividends, rights	Included in A1/D1	—

Category D MVP subtotal: ₹12,956 – ₹25,920/month (without optional D2).

Vendors: TrueData Corporate | <https://site.financialmodelingprep.com/pricing-plans> |
<https://www.simfin.com/en/prices/>

Category E — India Compliance & Shareholding Data

Purpose: Promoter pledge gate (P4) — pass if pledge < 10%, reject if > 15%.

#	API / Vendor	Purpose	Requirement s	MVP Monthly (₹)	One-Time (₹)
E1	TrueData getSHPListBy Company / getShpDetail ById	Promoter holding and pledge %	Quarterly shareholding pattern; historical SHP for backtest; symbol-level	Included in D1	—

			pledge field		
E2	Trendlyne API (fallback only)	Independent pledge verification if TrueData SHP incomplete	Current + historical pledge %; quarterly updates; NSE/BSE coverage	0 – 5,000 (if needed)	—

Category E subtotal: ₹0 – ₹5,000/month.

Category F — News & Document Ingestion (RAG)

Purpose: Feed the RAG pipeline for AI research panel — filings, news, announcements (not used in scoring).

#	API / Vendor	Purpose	Requirements	MVP Monthly (₹)	One-Time (₹)
F1	SEC EDGAR API	US 10-K, 10-Q, 8-K filing download	Free public API; CIK-to-ticker mapping; daily incremental ingest; PDF/HTML storage for RAG chunks	Free	—
F2	TrueData getAnnouncementFile / getAnnouncementsList	India corporate filing PDFs for RAG	Symbol-filtered announcements; file download by ID; daily ingest job	Included in D1	—
F3	Finnhub All-In-One	Company news feed (India + US)	300 calls/min; symbol-tagged news; daily RAG refresh; \$50/month plan	\$50 / ₹4,200	—
F4	FMP Market News API	US press releases and stock news	Included in FMP Premium/Ultimate	Included in D3	—

			subscription; symbol- filtered headlines		
F5	NewsAPI.org Business (Production only)	Broader news coverage for RAG at scale	250,000 requests/mo nth; commercial license; 5- year article search; CORS enabled	\$449 / ₹37,716 (Production)	—

Category F MVP subtotal: ₹4,200/month. Production: ₹4,200 – ₹41,916/month.

Category G — AI / LLM / RAG (Gemini)

Purpose: Research Q&A, filing summaries, score explanations — strictly NOT used for scoring or signals.

#	API / Vendor	Purpose	Requirements	MVP Monthly (₹)	One-Time (₹)
G1	Google Gemini 2.5 Flash API	Chat completion — “Why did X score 7/10?”	Paid tier; ~50 –200 queries/day; \$0.30/1M input + \$2.50/1M output tokens; no grounding in scoring path	1,500 – 5,000	—
G2	Google Gemini Embedding API (gemini- embedding-0 01)	RAG vector embeddings for filing chunks	\$0.15/1M tokens; batch API \$0.075/1M; store vectors in MongoDB; initial corpus embed + daily delta	500 – 2,000	—
G3	MongoDB Atlas Vector Search	Vector storage & similarity search	Self-hosted MongoDB — no external API fee if on existing cluster; Atlas	0 (self-hosted)	—

			Vector optional add-on		
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Category G subtotal: ₹2,000 – ₹7,000/month.

Vendor: <https://ai.google.dev/gemini-api/docs/pricing>

4. Scoring Point → API Mapping

Score Point	Description	Primary API Source
P1	Capital Velocity (ROE, ROIC, turnover)	D1 TrueData Corporate + D3 FMP + D5 SimFin
P2	Altman Z + Beneish M	D1, D3, D5 — computed in Laravel from statements
P3	Sloan Ratio	D1, D3, D5 — cash flow vs net income
P4	DSO + promoter pledge	D1, D3 + E1 TrueData SHP
P5	CAGR + PEG + FCFF/FCFE	D1, D3, D5 + A/B price data
P6	VPVR / HVN support	A1, A3, B1 — OHLCV volume history
P7	Mansfield RS vs index	C1 Nifty 50 + C2 S&P 500
P8	VCP — ATR/Close percentile	A1, A3, B1 — daily/minute bars
P9	Delivery % / block trades	A2/A3 DelVolume + B2 Polygon trades
P10	ADX trend maturity	A1, B1 — computed in Laravel from OHLCV

5. Consolidated Budget — MVP Scenario

Pilot: ~50 India + ~50 US symbols | Internal use | No client redistribution

Category	APIs Included	Monthly Low (₹)	Monthly High (₹)	One-Time (₹)
A — India market	A1–A4	10,000	25,000	50,000 – 1,50,000
B — US market	B1–B3	16,716	16,716	0
C — Indices	C1–C2	0	0	0
D — Fundamentals	D1, D3, D5, D6	12,956	25,920	0
E — India pledge	E1 (+ E2 if needed)	0	5,000	0
F — News / RAG	F1–F4	4,200	4,200	0

G — Gemini AI	G1-G2	2,000	7,000	0
TOTAL	—	45,872	83,836	50,000 – 1,50,000

MVP Year 1 Summary	Low (₹)	High (₹)
Monthly recurring × 12	5,50,464	10,06,032
One-time backfill	50,000	1,50,000
Subtotal (before GST)	6,00,464	11,56,032
GST @ 18% (est.)	1,08,084	2,08,086
GRAND TOTAL YEAR 1	7,08,548 (~₹7.1 Lakh)	13,64,118 (~₹13.6 Lakh)

6. Consolidated Budget — Production Scenario

Scale: ~200 symbols per market | Client-facing dashboard | Redistribution license likely required

Category	APIs Included	Monthly Low (₹)	Monthly High (₹)	One-Time (₹)
A — India market	A1-A5	35,000	80,000	1,50,000 – 4,00,000
B — US market	B1-B4	16,716	25,000	0
C — Indices	C1-C2	0	0	0
D — Fundamentals	D1-D5	35,000	55,000	0
E — India pledge	E1-E2	0	8,000	0
F — News / RAG	F1-F5	4,200	41,916	0
G — Gemini AI	G1-G2	5,000	15,000	0
TOTAL	—	95,916	2,25,000	1,50,000 – 4,00,000

Production Year 1 Summary	Low (₹)	High (₹)
Monthly recurring × 12	11,50,992	27,00,000
One-time backfill	1,50,000	4,00,000
Subtotal (before GST)	13,00,992	31,00,000
GST @ 18% (est.)	2,34,179	5,58,000
GRAND TOTAL YEAR 1	15,35,171 (~₹15.4 Lakh)	36,58,000 (~₹36.6 Lakh)

7. Recommended Approval Path

1. Approve MVP Low budget (~₹7.1 Lakh Year 1 incl. GST) for Phase 1 pilot.
2. Procure TrueData trial (10 days free) + Polygon Developer trial before full commit.

3. Confirm 20Y backfill depth and delivery % coverage with TrueData (historicaldata@truedata.in).
4. Confirm redistribution requirements if dashboard will be client-facing (Category A5).
5. Re-submit Production budget for approval after 90-day MVP validation.

8. Exclusions (Not in This Budget)

- Category H — Email/SMS alert APIs (Resend, Twilio, MSG91).
- Category J — Broker/trading APIs (Kite, IBKR, Alpaca Trading).
- Infrastructure — AWS/R2, VPS hosting, Sentry (optional, separate infra budget).
- Development team cost, Laravel hosting, domain, SSL.
- US market data display license beyond Polygon non-professional tier.

9. Client Approval

Approval Item	Approved (Yes/No)	Approved Budget (₹)	Sign / Date
MVP Scenario — Year 1 API budget			
Production Scenario — Year 1 API budget (deferred)			
TrueData as India data vendor			
Polygon.io as US data vendor			
Gemini as AI/RAG provider			

Authorised Signatory: _____ Date: _____

Client Name / Organisation: _____

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